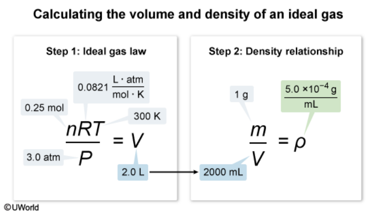
Assume that helium behaves as an ideal gas. What is the estimated density of 1.0 g of helium gas at a temperature of 27 °C and a pressure of 3.0 atm? (Note: Use R = 0.0821 L·atm·mol^{-1}·K^{-1})

- A. 1.3×10^{-4} g/mL [17%]
- B. 5.0×10^{-4} g/mL [52%]
- C. 1.0×10^{-3} g/mL [17%]
- D. 1.4×10^{-3} g/mL [13%]

Correct
51% answered correctly

Explanation:



An **ideal gas** is a hypothetical gas with well-behaved (idealized) characteristics used to make a **simplified model** to predict the behavior of **real gases**. Following this model, the behavior of a gas can be described by the **ideal gas law**:

$$P \quad V \quad = \quad n \quad R \quad T$$

where P is the pressure, V is the volume occupied, n is the number of moles, T is the absolute temperature (in Kelvin), and R is a constant.

If both the mass of a gas sample and the volume that it occupies are known, the gas **density *ρ*** can be evaluated as the **mass *m*** per unit **volume *V***:

$$\rho = \frac{m}{V}$$

In this question, the volume occupied by the helium gas sample under the given conditions must be calculated before the density can be estimated. Use of the ideal gas law requires that the mass of the 1.0-g sample of helium gas be converted into moles using the molar mass of helium:

$$1.0 \quad \cancel{\text{g He}} \quad \times$$

After conversion of the temperature from Celsius to Kelvin (27 °C + 273 = 300 K) and application of the ideal gas law (with R rounded for estimation), the volume occupied by the helium gas is:

$$V \quad = \quad \frac{n \quad R \quad T}{P}$$

Dividing the mass of the gas by the calculated volume (expressed in milliliters) gives an estimated density of:

$$\rho \quad = \quad \frac{m}{V} \quad =$$

(Choice A) 1.3×10^{-4} is obtained either by not converting the grams of helium to moles of helium when calculating the gas volume or by dividing the moles of helium, rather than the mass of helium, by the correctly calculated gas volume.

(Choice C) 1.0×10^{-3} is calculated by incorrectly using a molar mass of 8.00 g/mol. Like all noble gases, He has a full valence shell and exists as a *monatomic* gas in its elemental state. Because the valence shell of He (ie, a 2*s* orbital) has a maximum capacity of 2 electrons, He does not form bonds with other atoms. As such, diatomic He_2(*g*) *does not form* and helium is *not* one of the common diatomic elements.

(Choice D) The mass of gas must be converted into moles and the temperature must be converted from Celsius to Kelvin when using the ideal gas law.

Educational objective:
An ideal gas is a hypothetical gas with well-behaved (idealized) characteristics used to make a simplified model to predict the behavior of real gases. The behavior of a gas can be described by the ideal gas law. Using a calculated gas volume and its mass, the gas density (mass per unit volume) can be evaluated.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402582
System:
Thermodynamics, Kinetics & Gas Laws